Department of Education
TVL
Oracle Database Programming NC III

Module 1

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What I Need to Know

Welcome!

Computer programming is the act of writing computer programs, which are a sequence of instructions written using a Computer Programming Language to perform a specified task by the computer.

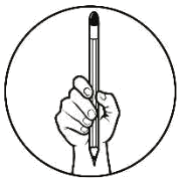
Computer Programming is fun and easy to learn provided you adopt a proper approach. This module attempts to cover the basics of computer programming using a simple and practical approach.

The module is divided into three lessons namely:

- Lesson 1 – Computer Programming - Overview
- Lesson 2 – Computer Programming – Environment and Basics
- Lesson 3 – Computer Programming – Basic Syntax

After going through this module, you are expected to:

1. Define Computer Programming.
2. Enumerate the different programming language.
3. Define the uses of Computer Programs.
4. Define algorithm and its importance in programming.
5. Enumerate the basic elements of Computer Programming.



What I Know

IRF Worksheet: What is Computer Programming?

Instruction: There are three (3) given boxes below – (I) – Stands for your Initial answer: Here, you will put your prior knowledge on the given question: What is Computer Programming?, (R) – Stands for Revise answer: Here, you will put your Revise answer after the lesson is read/discussed, (F) – Stands for Final answer; Here you will put your final answer after the final assessment is taken/given. For now, please answer the INITIAL ANSWER part.

INITIAL ANSWER
REVISED ANSWER
FINAL ANSWER

Lesson 1 Computer Programming Overview

I believe you have used a running computer program; it might be on your smartphone, tablets, laptops or personal computers. Even wonder how it works? Let's have an overview of it.

What's In

A pre-requisite for this overview lesson of programming is that you have used any word processing application or enough amount of knowledge on the basic parts of the computer such as mouse, keyboard, monitor. Etc.

What's New

Before the lesson, lets start identifying what are programming languages. From your prior knowledge, which do you think are Programming Languages, encircle them.

WPS	Java	Windows Store
MS Word	C	Windows Store
MS Excel	C++	Adobe Photoshop
Internet Explorer	Facebook	PHP
Google Chrome	Zoom	SQL

What is It

Before getting into computer programming, let us first understand computer programs and what they do.

A computer program is a sequence of instructions written using a Computer Programming Language to perform a specified task by the computer.

The two important terms that we have used in the above definition are −

- Sequence of instructions
- Computer Programming Language

To understand these terms, consider a situation when someone asks you about how to go to a nearby KFC. What exactly do you do to tell him the way to go to KFC?

You will use Human Language to tell the way to go to KFC, something as follows −

First go straight, after half kilometer, take left from the red light and then drive around one kilometer and you will find KFC at the right.

Here, you have used English Language to give several steps to be taken to reach KFC. If they are followed in the following sequence, then you will reach KFC −

1. Go straight
2. Drive half kilometer
3. Take left
4. Drive around one kilometer
5. Search for KFC at your right side

Now, try to map the situation with a computer program. The above sequence of instructions is actually a **Human Program** written in **English Language**, which instructs on how to reach KFC from a given starting point. This same sequence could have been given in Spanish, Hindi, Arabic, or any other human language, provided the person seeking direction knows any of these languages.

Now, let's go back and try to understand a computer program, which is a sequence of instructions written in a Computer Language to perform a specified task by the computer. Following is a simple program written in **Python** programming Language −

```
print "Hello, World!"
```

The above computer program instructs the computer to print "Hello, World!" on the computer screen.

- A computer program is also called a **computer software**, which can range from two lines to millions of lines of instructions.
- Computer program instructions are also called program source code and **computer programming** is also called **program coding**.
- A computer without a computer program is just a dump box; it is programs that make computers active.

As we have developed so many languages to communicate among ourselves, computer scientists have developed several computer-programming languages to provide instructions to the computer (i.e., to write computer programs). We will see several computer programming languages in the subsequent chapters.

Introduction to Computer Programming

If you understood what a **computer program** is, then we will say: *the act of writing computer programs is called computer programming.*

As we mentioned earlier, there are hundreds of programming languages, which can be used to write computer programs and following are a few of them −

- Java
- C
- C++
- Python
- PHP
- Perl
- Ruby

Uses of Computer Programs

Today computer programs are being used in almost every field, household, agriculture, medical, entertainment, defense, communication, etc. Listed below are a few applications of computer programs −

- MS Word, MS Excel, Adobe Photoshop, Internet Explorer, Chrome, etc., are examples of computer programs.
- Computer programs are being used to develop graphics and special effects in movie making.
- Computer programs are being used to perform Ultrasounds, X-Rays, and other medical examinations.
- Computer programs are being used in our mobile phones for SMS, Chat, and voice communication.

Algorithm

From programming point of view, an **algorithm** is a step-by-step procedure to resolve any problem. An algorithm is an effective method expressed as a finite set of well-defined instructions.

Thus, a computer programmer lists down all the steps required to resolve a problem before writing the actual code. Following is a simple example of an algorithm to find out the largest number from a given list of numbers −

1. Get a list of numbers L_1, L_2, L_3 ..... L_N
2. Assume L_1 is the largest, Largest = L_1
3. Take next number L_i from the list and do the following
4. If Largest is less than L_i
5. Largest = L_i
6. If L_i is last number from the list then
7. Print value stored in Largest and come out
8. Else repeat same process starting from step 3

The above algorithm has been written in a crude way to help beginners understand the concept. You will come across more standardized ways of writing computer algorithms as you move on to advanced levels of computer programming.

What's More

Search for more Programming Languages available nowadays and compare them according to their uses.

_____

What I Have Learned

Computer Programming is the sequence of instructions made by programmers to perform specific task.

Programming Languages vary on different algorithms to write on specific programs/system.

Algorithm is the sequence of procedure of different programming languages to resolve any problem. It is an effective way expressed as a finite set of well-defined instructions.

What I Can Do

At the beginning of this lesson, you have answered the IRF Worksheet and answered the INITIAL ANSWER part, this time you will answer the REVISED ANSWER part as you have finished the lesson and reflect on your answer.

IRF Worksheet: What is Computer Programming?

INITIAL ANSWER

REVISED ANSWER
FINAL ANSWER

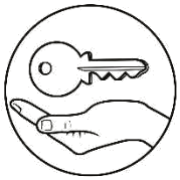
Assessment

PART I: Which of the following are considered as Programming Languages? Put a check mark on each of the numbers.

1. C++
2. Twitter
3. PHP
4. Facebook
5. SQL
6. Word
7. Java
8. Canva
9. Python
10. USB

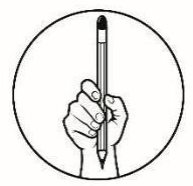
PART II: Short Answer: Using your own learning, write your brief description on the given terms below.

1. Computer Programming
2. Programming Language
3. Algorithm
4. Computer Programmer
5. Sequence Instructions



Answer Key

	10 Yes	
	9 Yes	
	8	
	7 Yes	
	6	
	5 Yes	SQL
	4	PHP
	3 Yes	C++
	2	C
	1 Yes	Java
	Part I:	
	Assessment	**What's New**



What I Know

From the previous lesson, we learned the essentials of Computer Programming that it is a set of instructions to perform specific task or to build a program. Also, algorithm is a set of step-by-step rules that governs a computer program. Some given Programming Languages are given like Java, PHP, SQL etc. Now, we will learn the relationship of English Language to Computer Programming and to Programming Languages.

Lesson 2 Computer Programming – Basics and Environment

We assume you are well aware of English Language, which is a well-known **Human Interface Language**. English has a predefined grammar, which needs to be followed to write English statements in a correct way. Likewise, most of the Human Interface Languages (Hindi, English, Spanish, French, etc.) are made of several elements like verbs, nouns, adjectives, adverbs, propositions, and conjunctions, etc. Similar to that is Computer Programming that will be discussed on this lesson.

What's In

On this lesson, you should be knowledgeable on the essentials of Computer Programming, Algorithms and the sample basic Programming Languages to be able to grasps the basics and environment of Computer Programming.

What's New

From your prior knowledge on the essentials in Computer Programming Overview, which do you think are the elements of a Programming Language?

Programming Environment	Basic Syntax	Data Types
Variables	Keywords	Basic Operators
Decision Making	Characters	Numbers

What is It

Similar to Human Interface Languages, Computer Programming Languages are also made of several elements. These basic elements include –

- **Programming Environment**
- **Basic Syntax**
- **Data Types**
- **Variables**
- **Keywords**
- **Basic Operators**
- **Decision Making**
- **Loops**
- **Numbers**
- **Characters**
- **Arrays**
- **Strings**
- **Functions**
- **File I/O**

First, we will try to understand the meaning of all these terms in general and then, we will see how these terms can be used in different programming languages.

This lesson has been designed to give you an idea about the following most popular programming languages −

- **C Programming**
- **Java Programming**
- **Python Programming**

A major part of the tutorial has been explained by taking C as programming language and then we have shown how similar concepts work in Java and Python. So after completion of this lesson, you will be quite familiar with these popular programming languages.

Environment

Though Environment Setup is not an element of any Programming Language, it is the first step to be followed before setting on to write a program.

When we say Environment Setup, it simply implies a base on top of which we can do our programming. Thus, we need to have the required software setup, i.e., installation on our PC which will be used to write computer programs, compile, and execute them.

Similarly, you will need the following setup to start with programming using any programming language.

- A text editor to create computer programs.
- A compiler to compile the programs into binary format.
- An interpreter to execute the programs directly.

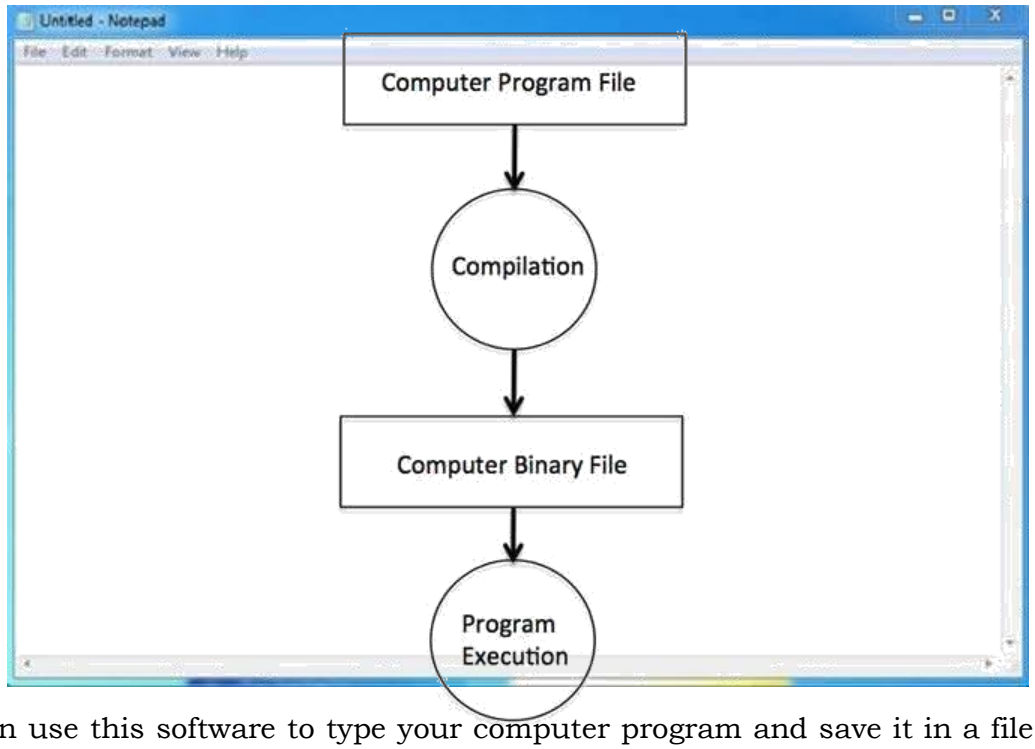
In case you don't have sufficient exposure to computers, you will not be able to set up either of these software.

Text Editor

A text editor is a software that is used to write computer programs. Your Windows machine must have a Notepad, which can be used to type programs. You can launch it by following these steps −

Start Icon → All Programs → Accessories → Notepad → Mouse Click on Notepad

It will launch Notepad with the following window −



You can use this software to type your computer program and save it in a file at any location. You can download and install other good editors like **Notepad++**, which is freely available.

If you are a Mac user, then you will have **TextEdit** or you can install some other commercial editor like **BBEdit** to start with.

Compiler

You write your computer program using your favorite programming language and save it in a text file called the program file.

Now let us try to get a little more detail on how the computer understands a program written by you using a programming language. Actually, the computer cannot understand your program directly given in the text format, so we need to convert this program in a binary format, which can be understood by the computer.

The conversion from text program to binary file is done by another software called Compiler and this process of conversion from text formatted program to binary format file is called program compilation. Finally, you can execute binary file to perform the programmed task.

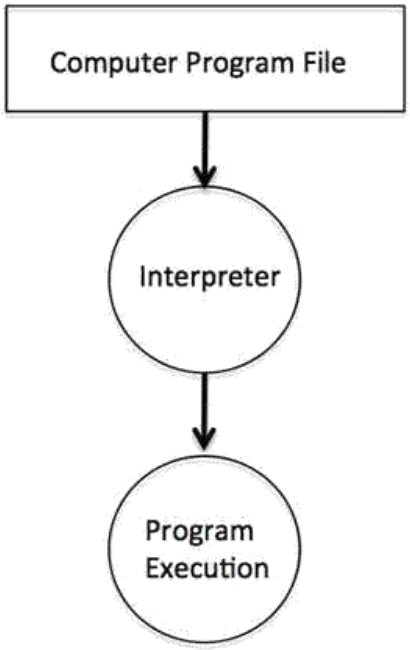
We are not going into the details of a compiler and the different phases of compilation. The following flow diagram gives an illustration of the process –

So, if you are going to write your program in any such language, which needs compilation like C, C++, Java and Pascal, etc., then you will need to install their compilers before you start programming.

Interpreter

We just discussed about compilers and the compilation process. Compilers are required in case you are going to write your program in a programming language that needs to be compiled into binary format before its execution.

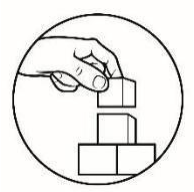
There are other programming languages such as Python, PHP, and Perl, which do not need any compilation into binary format, rather an interpreter can be used to read such programs line by line and execute them directly without any further conversion.



So, if you are going to write your programs in PHP, Python, Perl, Ruby, etc., then you will need to install their interpreters before you start programming.

Online Compilation

If you are not able to set up any editor, compiler, or interpreter on your machine, then *tutorialspoint.com* provides a facility to compile and run almost all the programs online with an ease of a single click.



What's More

You can search for the other Programming Language Elements such as Data Types and Variables, and how does it different from one language to the other

What I Have Learned

The basic elements of Programming Languages:

- Programming Environment
- Basic Syntax
- Data Types
- Variables
- Keywords
- Basic Operators
- Decision Making
- Loops
- Numbers
- Characters
- Arrays
- Strings
- Functions
- File I/O

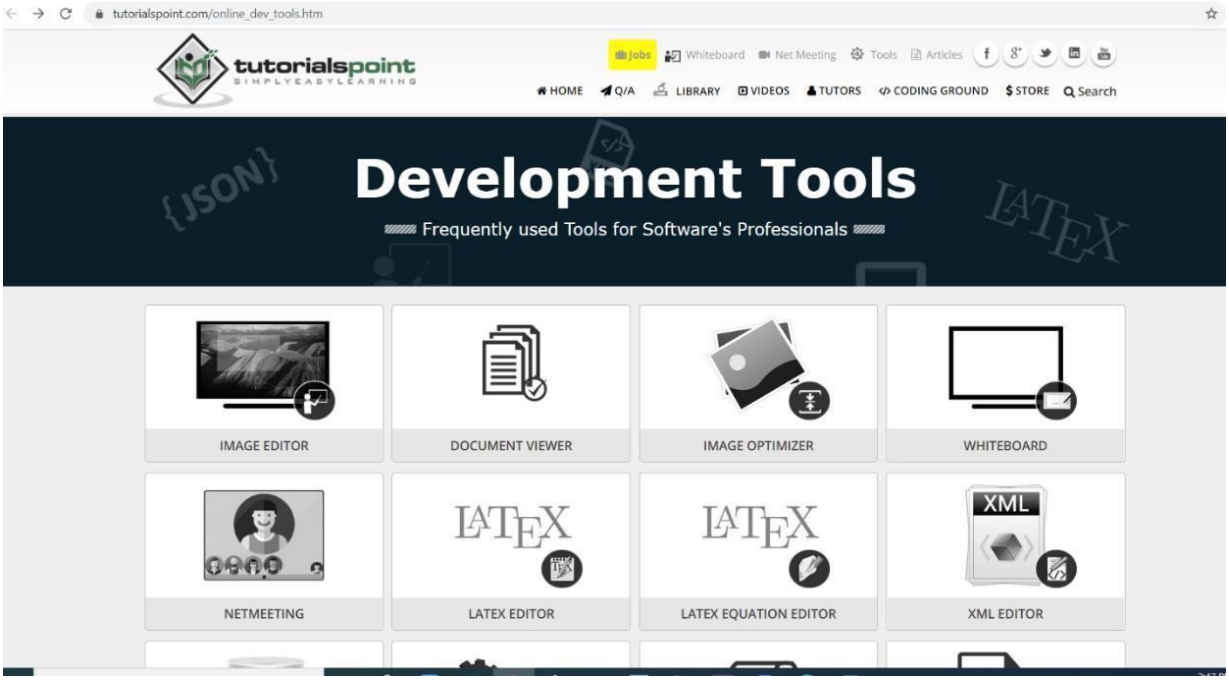
Three Programming Environment Essentials:

- A text editor to create computer programs.
- A compiler to compile the programs into binary format.
- An interpreter to execute the programs directly.

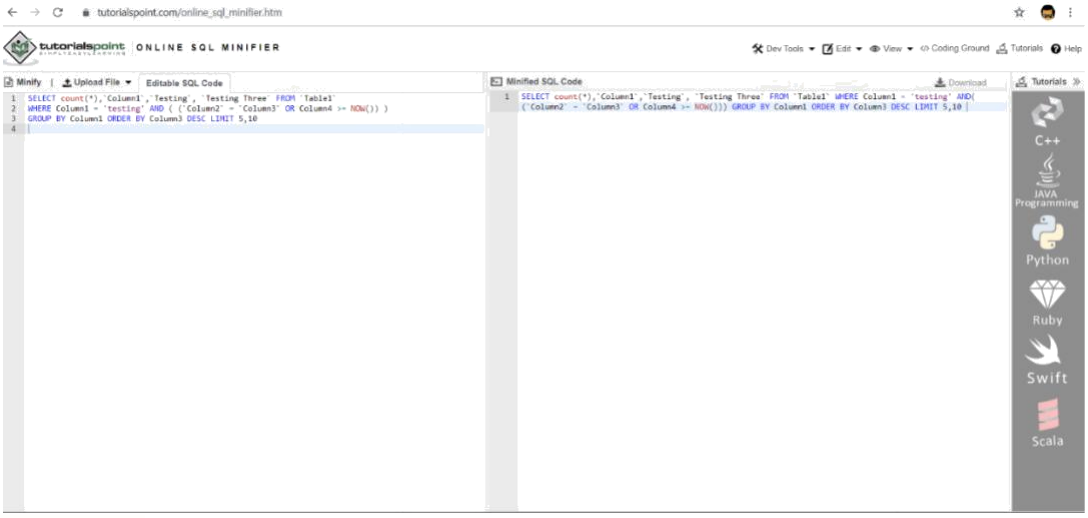
What I Can Do

Now, to try on different Programming Language and its basics: Environment, Algorithm and Basic Syntax, you can check Tutorialspoint Website with this link: https://www.tutorialspoint.com/online_dev_tools.htm and choose of any programming language that you wanted to learn the basics. For example: SQL (see picture below).

Tutorialspoint Development Tools view.



Tutorialspoint Development Tools -SQL Programming view.



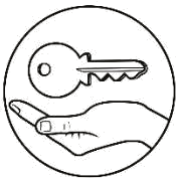
Assessment

Part I: Which of the following are the Elements of Programming Language.

Functions
Strings
Arrays
Characters
Numbers
Keywords
Variables
Basic Syntax
Decision Making
Computer Peripherals
Motherboard
Input/Output Device

Part II:
Define the three terms in your own words as you learned it from the lesson.

A. Text Editor
B. Compiler
C. Interpreter



Answer Key

	Decision Making Basic Syntax Variables Keywords Numbers Characters Arrays Strings Functions Part I: **Assessment**	Characters Numbers Loops Decision Making Basic Operators Keywords Variables Data Types Basic Syntax Programming Environment **What's New**

Lesson 3	Computer Programming – Basics Syntax

Coding is basically the computer language used to develop apps, websites, and software. Without it, we'd have none of the most popular technology we've come to rely on such as Facebook, our smartphones, the browser we choose to view our favorite blogs, or even the blogs themselves. It all runs on code.

What's In

On this lesson, you should be knowledgeable on the basic syntax of selected Programming Language such as Java and Python and learn the semantics and algorithm of each.

What's New

From your prior knowledge on the essentials in Computer Programming – Basic Environment, there are three main environment essentials to start a project. The three main environment are listed below, define each term and give its use on programming.

1. TEXT EDITOR
2. COMPILER
3. INTERPRETER

What is It

Let's start with a little coding, which will really make you a computer programmer. We are going to write a single-line computer program to write **Hello, World!** on your screen. Let's see how it can be written using different programming languages.

Hello World Program in C

Try the following example using our online compiler option available at www.compileonline.com.

For most of the examples given in this tutorial, you will find a **Try it** option in our website code sections at the top right corner that will take you to the online compiler.

Try to change the content inside printf(), i.e., type anything in place of **Hello World!** and then check its result. It just prints whatever you keep inside the two double quotes.

```
#include <stdio.h>

int main() {
   /* printf() function to write Hello, World! */
   printf( "Hello, World!" );
}
```

which produces the following result −

Hello, World!

This little Hello World program will help us understand various basic concepts related to C Programming.

Program Entry Point

For now, just forget about the **#include <stdio.h>** statement, but keep a note that you have to put this statement at the top of a C program.

Every C program starts with main(), which is called the main function, and then it is followed by a left curly brace. The rest of the program instruction is written in between and finally a right curly brace ends the program.

The coding part inside these two curly braces is called the program body. The left curly brace can be in the same line as main(){ or in the next line like it has been mentioned in the above program.

Functions

Functions are small units of programs and they are used to carry out a specific task. For example, the above program makes use of two functions: **main()** and **printf()**. Here, the function main() provides the entry point for the program execution and the other function printf() is being used to print an information on the computer screen.

You can write your own functions which we will see in a separate chapter, but C programming itself provides various built-in functions like main(), printf(), etc., which we can use in our programs based on our requirement.

Some of the programming languages use the word **sub-routine** instead of function, but their functionality is more or less the same.

Comments

A C program can have statements enclosed inside **/*.... */**. Such statements are called comments and these comments are used to make the programs user friendly

and easy to understand. The good thing about comments is that they are completely ignored by compilers and interpreters. So you can use whatever language you want to write your comments.

Whitespaces

When we write a program using any programming language, we use various printable characters to prepare programming statements. These printable characters are **a, b, c,.....z, A, B, C,.....Z, 1, 2, 3,. 0, !, @, #, $, %, ^, &, *, (,), -, _, +, =, \, |, {, }, [,], :, ;, <, >, ?, /, \, ~. `. ", '**. Hope I'm not missing any printable characters from your keyboard.

Apart from these characters, there are some characters which we use very frequently but they are invisible in your program and these characters are spaces, tabs (\t), new lines(\n). These characters are called **whitespaces**.

These three important whitespace characters are common in all the programming languages and they remain invisible in your text document −

Whitespace	Explanation	Representation
New Line	To create a new line	\n
Tab	To create a tab.	\t
Space	To create a space.	empty space

A line containing only whitespace, possibly with a comment, is known as a blank line, and a C compiler totally ignores it. Whitespace is the term used in C to describe blanks, tabs, newline characters, and comments. So you can write **printf("Hello, World!");** as shown below. Here all the created spaces around "Hello, World!" are useless and the compiler will ignore them at the time of compilation.

```
#include <stdio.h>

int main() {

   /* printf() function to write Hello, World! */

   printf(    "Hello, World!"       );

}
```

which produces the following result −

Hello, World!

If we make all these whitespace characters visible, then the above program will look like this and you will not be able to compile it −

```
#include <stdio.h>\n


\n
int main()\n
{
   \n
   \t/* printf() function to write Hello, World! */
   \n
   \tprintf(\t"Hello, World!"\t);\n
   \n
}\n
```

Semicolons

Every individual statement in a C Program must be ended with a semicolon (**;**), for example, if you want to write "Hello, World!" twice, then it will be written as follows –

```
#include <stdio.h>

int main() {
   /* printf() function to write Hello, World! */
   printf( "Hello, World!\n" ); printf( "Hello,
   World!" );
}
```

This program will produce the following result –

Hello, World!
Hello, World!

Here, we are using a new line character **\n** in the first printf() function to create a new line. Let us see what happens if we do not use this new line character –

```
#include <stdio.h>

int main() {
   /* printf() function to write Hello, World! */
   printf( "Hello, World!" ); printf( "Hello,
   World!" );
}
```

This program will produce the following result –

Hello, World! Hello, World!

Program Explanation

Let us understand how the above C program works. First of all, the above program is converted into a binary format using C compiler. So let's put this code in test.c file and compile it as follows –

$gcc test.c -o demo

If there is any grammatical error (Syntax errors in computer terminologies), then we fix it before converting it into binary format. If everything goes fine, then it produces a binary file called **demo**. Finally, we execute the produced binary demo as follows –

$./demo

which produces the following result –

Hello, World!

Here, when we execute the binary **a.out** file, the computer enters inside the program starting from main() and encounters a printf() statement. Keep a note that the line inside /* */ is a comment and it is filtered at the time of compilation. So printf() function instructs the computer to print the given line at the computer screen. Finally, it encounters a right curly brace which indicates the end of main() function and exits the program.

Syntax Error

If you do not follow the rules defined by the programing language, then at the time of compilation, you will get syntax errors and the program will not be compiled. From syntax point of view, even a single dot or comma or a single semicolon matters and you should take care of such small syntax as well. In the following example, we have skipped a semicolon, let's try to compile the program –

```
#include <stdio.h>

main() {
  printf("Hello, World!")
}
```

This program will produce the following result –

main.c: In function 'main':

main.c:7:1: error: expected ';' before '}' token

}

^

So the bottom-line is that if you are not following proper syntax defined by the programming language in your program, then you will get syntax errors. Before attempting another compilation, you will need to fix them and then proceed.

Hello World Program in Java

Following is the equivalent program written in Java. This program will also produce the same result **Hello, World!**.

```
public class HelloWorld {
  public static void main(String []args) {
    /* println() function to write Hello, World! */
    System.out.println("Hello, World!");
  }
}
```

which produces the following result −

Hello, World!
Hello World Program in Python

Following is the equivalent program written in Python. This program will also produce the same result **Hello, World!**.

```
# print function to write Hello, World!
*/ print "Hello, World!"
```

which produces the following result −

Hello, World!

Hope you noted that for C and Java examples, first we are compiling the programs and then executing the produced binaries, but in Python program, we are directly executing it. As we explained in the previous chapter, Python is an interpreted language and it does not need an intermediate step called compilation.

Python does not require a semicolon (;) to terminate a statement, rather a new line always means termination of the statement.

What's More

From the lesson, basic display of "HELLO WORLD" from C Programming and Python were shown. Now, you can search for other Programming Language to display "HELLO WORLD" like JAVA, C++ and the others. Afterwards, compare each programming language and distinguish their differences and similarities, especially on semantics and algorithms.

What I Have Learned

I. The basic unit of a Program
- Program Entry Point
- Functions
- Comments
- Whitespaces
- Semi Colon
- Program Explanation
- Syntax Error

II. Displaying basic "HELLO WORLD" in C Programming and Python; the similarities and differences of their algorithms.

What I Can Do

Now, lets try displaying HELLO WORLD in a Programming Language with LIVE compiler. Go to: https://www.tutorialspoint.com/compile_c_online.php and try to display YOUR FULL NAME: GIVEN NAME, MIDDLE NAME and SURNAME using C PROGRAMMING, please refer to the given example on the lesson.

TIP: ***printf()*** is a function in C Programming to display any text on the screen.

Assessment

From this lesson, you've learned the essentials in Programming, the Basic Environment and the Basic Syntax of selected Programming Language, Now lets try your learning and do a task.

Your main task is to display the following details in any programming language which you think you can use the most:

- NAME:
- AGE:
- GENDER:
- BIRTHDAY:
- SCHOOL:
- GRADE LEVEL:
- SECTION:
- WHAT IS PROGRAMMING FOR YOU:

All you have to do is to fill-up the given information and display it using a Programming Language.

TIP: You can use online compiler on TUTORIALSPOINT.COM or go to: https://www.tutorialspoint.com/online_dev_tools.htm or if your using C Programming: https://www.tutorialspoint.com/compile_c_online.php or if your using Java: https://www.tutorialspoint.com/online_java_formatter.htm.

After coding, submit the SOURCE CODE and attach the screenshot of the COMPILED PROGRAM to: ryanjeff.mercado@deped.gov.ph or

ryanjeffmercado@gmail.com or on FB-messenger: Mercado, Ryan Jeff. Please include your Full Name upon submission.

Rubric to check your output.

Program (50 pts)	**(Excellent)**	**(Good)**	**(Fair)**	**(Poor)**
Program execution	Program executes correctly with no syntax or runtime errors (9-10)	Program executes fairly with 1 or 2 syntax or runtime errors (6-8)	Program executes with a minor (easily fixed error) (2-3)	Program does not execute (0-1)
Correct output	Program displays correct output with no errors (9-10)	Output has minor errors (6-8)	Output has multiple errors (3-5)	Output is incorrect (0-2)
Design of output	Program displays more than expected (7-8)	Program displays minimally expected output (5-6)	Program does not display the required output (3-4)	Output is poorly designed (0-2)
Design of logic	Program is logically well designed (9-10)	Program has slight logic errors that do no significantly affect the results (6-8)	Program has significant logic errors (3-5)	Program is incorrect (0-2)
Standards	Program is stylistically well designed (6-7)	Few inappropriate design choices (i.e. poor variable names, improper indentation) (4-5)	Several inappropriate design choices (i.e. poor variable names, improper indentation) (2-3)	Program is poorly written (0-1)
Documentation	Program is well documented, screenshot is properly attached (5)	Missing one required comment (4)	Missing two or more required comments (2-3)	Most or all documentation missing (0-1)

Answer Key

See Rubric

Assessment

References

Tutorialspoint, Computer Programming-Home 2020:
https://www.tutorialspoint.com/computer_programming/index.htm

Tutorialspoint, Computer Programming-Overview 2020:
https://www.tutorialspoint.com/computer_programming/computer_programming_overview.htm

Tutorialspoint, Computer Programming-Basics 2020:
https://www.tutorialspoint.com/computer_programming/computer_programming_basics.htm

Tutorialspoint, Computer Programming-Environment 2020:
https://www.tutorialspoint.com/computer_programming/computer_programming_environment.ht m

Grading Rubric: http://csc.columbusstate.edu/summers/NOTES/1301/Grading-Rubric-Programs.htm